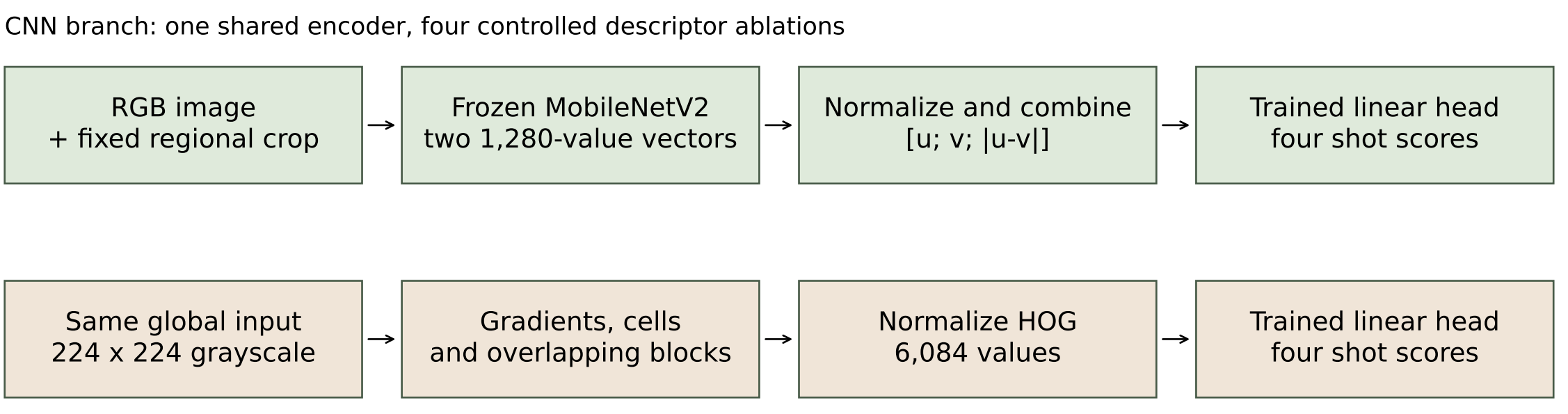
CNN branch: one shared encoder, four controlled descriptor ablations
RGB image
+ fixed regional crop
Frozen MobileNetV2
two 1,280-value vectors
Normalize and combine
[u; v; |u-v|]
Trained linear head
four shot scores
Same global input
224 x 224 grayscale
Gradients, cells
and overlapping blocks
Normalize HOG
6,084 values
Trained linear head
four shot scores
HOG branch: deterministic features; identical labels, visual groups and train/validation/test split